E1 Basic trigonometry

Name:

Class:

Date:

Time: **164 minutes**

Marks: **136 marks**

Comments:

Q1.

(a) Solve the equation $\tan x = -\frac{1}{3}$, giving all the values of x in the interval $0 < x < 2\pi$ in radians to two decimal places.

(3)

(b) Show that the equation

$$3\sec^2 x = 5(\tan x + 1)$$

can be written in the form $3\tan^2 x - 5\tan x - 2 = 0$.

(1)

(c) Hence, or otherwise, solve the equation

$$3\sec^2 x = 5(\tan x + 1)$$

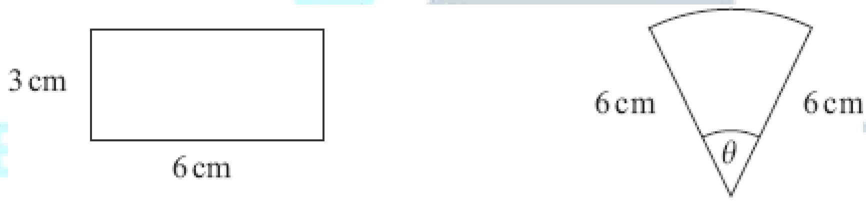
giving all the values of x in the interval $0 < x < 2\pi$ in radians to two decimal places.

(4)

(Total 8 marks)

Q2.

The diagrams show a rectangle of length 6 cm and width 3 cm, and a sector of a circle of radius 6 cm and angle θ radians.



The area of the rectangle is twice the area of the sector.

(a) Show that $\theta = 0.5$.

(3)

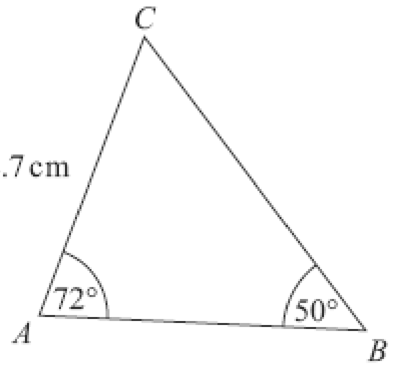
(b) Find the perimeter of the sector.

(3)

(Total 6 marks)

Q3.

The diagram shows a triangle ABC. The length of AC is 18.7 cm, and the sizes of angles BAC and ABC are 72° and 50° respectively.



(a) Show that the length of BC = 23.2 cm, correct to the nearest 0.1 cm.

(3)

(b) Calculate the area of triangle ABC, giving your answer to the nearest cm^2.

(3)

(Total 6 marks)

Q4.

(a) Solve the equation $\cot x = 2$, giving all values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(2)

(b) Show that the equation $\operatorname{cosec}^2 x = \dfrac{3\cot x + 4}{2}$ can be written as

$$2\cot^2 x - 3\cot x - 2 = 0$$

(2)

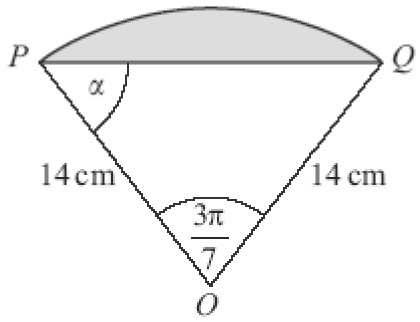
(c) Solve the equation $\operatorname{cosec}^2 x = \dfrac{3\cot x + 4}{2}$, giving all values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(4)

(Total 8 marks)

Q5.

The diagram shows a shaded segment of a circle with centre O and radius 14 cm, where PQ is a chord of the circle.



In triangle OPQ, angle $POQ = \frac{3\pi}{7}$ radians and angle $OPQ = \alpha$ radians.

(a) Find the length of the arc PQ, giving your answer as a multiple of π.

(2)

(b) Find α in terms of π.

(2)

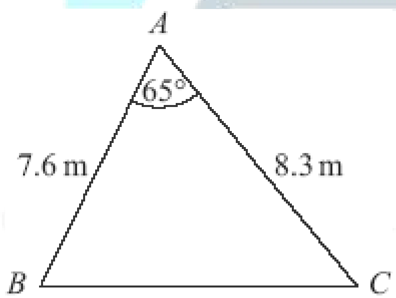
(c) Find the **perimeter** of the shaded segment, giving your answer to three significant figures.

(2)

(Total 6 marks)

Q6.

The diagram shows a triangle ABC.



The size of angle BAC is 65°, and the lengths of AB and AC are 7.6 m and 8.3 m respectively.

(a) Show that the length of BC is 8.56 m, correct to three significant figures.

(3)

(b) Calculate the area of triangle ABC, giving your answer in m² to three significant figures.

(2)

(c) The perpendicular from A to BC meets BC at the point D.

Calculate the length of AD, giving your answer to the nearest 0.1 m.

(3)
(Total 8 marks)

Q7.

(a) Solve the equation sec x = 3, giving the values of x in radians to two decimal places in the interval $0 \leq x < 2\pi$.

(3)

(b) Show that the equation $\tan^2 x = 2 \sec x + 2$ can be written as $\sec^2 x - 2 \sec x - 3 = 0$.

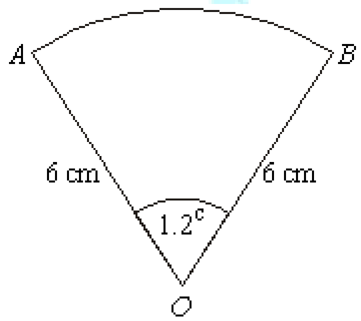
(2)

(c) Solve the equation $\tan^2 x = 2 \sec x + 2$, giving the values of x in radians to two decimal places in the interval $0 \leq x < 2\pi$.

(4)
(Total 9 marks)

Q8.

The diagram shows a sector OAB of a circle with centre O.



The radius of the circle is 6 cm and the angle AOB is 1.2 radians.

(a) Find the area of the sector OAB.

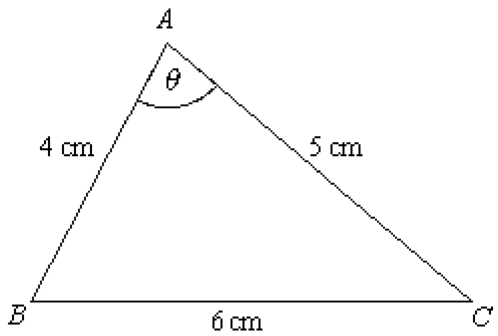
(2)

(b) Find the perimeter of the sector OAB.

(3)
(Total 5 marks)

Q9.

The triangle ABC, shown in the diagram, is such that BC = 6 cm, AC = 5 cm and AB = 4 cm. The angle BAC is θ.



(a) Use the cosine rule to show that $\cos\theta = \frac{1}{8}$.

(3)

(b) Hence use a trigonometrical identity to show that $\sin\theta = \frac{3\sqrt{7}}{8}$.

(3)

(c) Hence find the area of the triangle ABC.

(2)

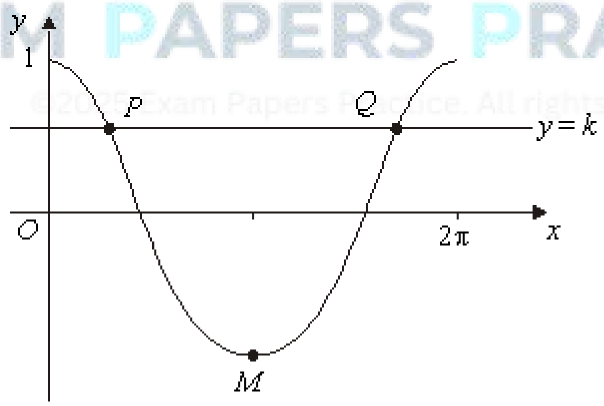
(Total 8 marks)

Q10.

(a) Solve the equation cos x = 0.3 in the interval $0 \le x \le 2\pi$, giving your answers in radians to three significant figures.

(3)

(b) The diagram shows the graph of $y = \cos x$ for $0 \le x \le 2\pi$, and the line $y = k$.



The line $y = k$ intersects the curve $y = \cos x$, $0 \le x \le 2\pi$, at the points P and Q. The point M is the minimum point of the curve.

(i) Write down the coordinates of the point M.

(2)

(ii) The x-coordinate of P is α.

Write down the x-coordinate of Q in terms of π and α.

(1)

(c) Describe the geometrical transformation that maps the graph of $y = \cos x$ onto the graph of $y = \cos 2x$.

(2)

(d) Solve the equation $\cos 2x = \cos \frac{4\pi}{5}$ in the interval $0 \leq x \leq 2\pi$, giving the values of x in terms of π.

(4)

(Total 12 marks)

Q11.

(a) (i) Show that the equation

$2 \cot^2 x + 5 \operatorname{cosec} x = 10$

can be written in the form $2 \operatorname{cosec}^2 x + 5 \operatorname{cosec} x - 12 = 0$.

(2)

(ii) Hence show that $\sin x = -\frac{1}{4}$ or $\sin x = \frac{2}{3}$.

(3)

(b) Hence, or otherwise, solve the equation

$2 \cot^2(\theta - 0.1) + 5 \operatorname{cosec}(\theta - 0.1) = 10$

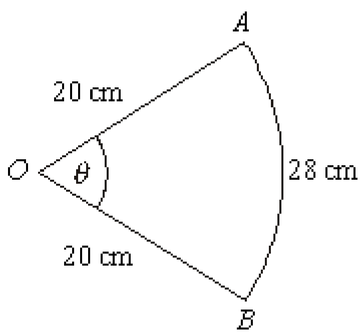
giving all values of θ in radians to two decimal places in the interval $-\pi < \theta < \pi$.

(3)

(Total 8 marks)

Q12.

The diagram shows a sector OAB of a circle with centre O and radius 20 cm. The angle between the radii OA and OB is θ radians.



The length of the arc AB is 28 cm.

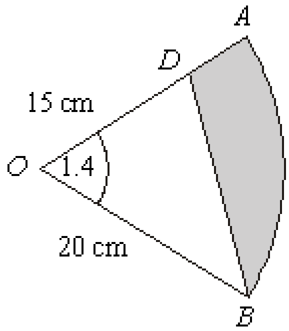
(a) Show that θ = 1.4.

(2)

(b) Find the area of the sector OAB.

(2)

(c) The point D lies on OA. The region bounded by the line BD, the line DA and the arc AB is shaded.



The length of OD is 15 cm.

(i) Find the area of the shaded region, giving your answer to three significant figures.

(3)

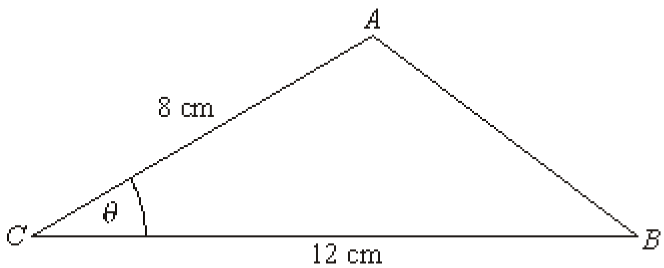
(ii) Use the cosine rule to calculate the length of BD, giving your answer to three significant figures.

(3)

(Total 10 marks)

Q13.

The triangle ABC, shown in the diagram, is such that AC = 8 cm, CB = 12 cm and angle ACB = θ radians.



The area of triangle ABC = 20 cm^2.

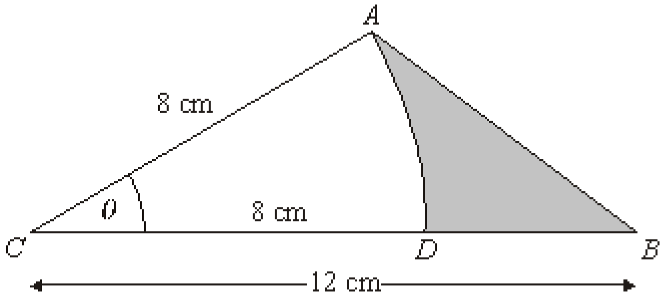
(a) Show that θ = 0.430 correct to three significant figures.

(3)

(b) Use the cosine rule to calculate the length of AB, giving your answer to two significant figures.

(3)

(c) The point D lies on CB such that AD is an arc of a circle centre C and radius 8 cm. The region bounded by the arc AD and the straight lines DB and AB is shaded in the diagram.



Calculate, to two significant figures:

(i) the length of the arc AD;

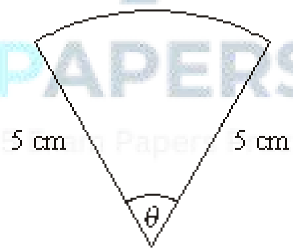
(2)

(ii) the area of the shaded region.

(3)
(Total 11 marks)

Q14.

The diagram shows a sector of a circle of radius 5 cm and angle θ radians.



The area of the sector is 8.1 cm^2.

(a) Show that $\theta = 0.648$.

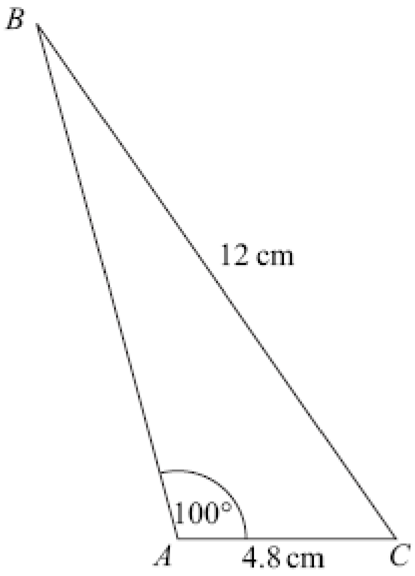
(2)

(b) Find the perimeter of the sector.

(3)
(Total 5 marks)

Q15.

The diagram shows a triangle ABC.



The lengths of AC and BC are 4.8 cm and 12 cm respectively.

The size of the angle BAC is 100°.

(a) Show that angle ABC = 23.2°, correct to the nearest 0.1°.

(3)

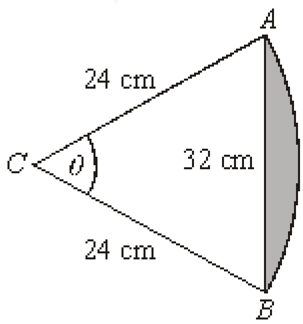
(b) Calculate the area of triangle ABC, giving your answer in cm^2 to three significant figures.

(3)

(Total 6 marks)

Q16.

The diagram shows a triangle ABC and the arc AB of a circle whose centre is C and whose radius is 24 cm.



The length of the side AB of the triangle is 32 cm. The size of the angle ACB is θ radians.

(a) Show that θ = 1.46 correct to three significant figures.

(3)

(b) Calculate the length of the arc AB to the nearest cm.

(2)

(c) (i) Calculate the area of the sector ABC to the nearest cm^2.

(2)

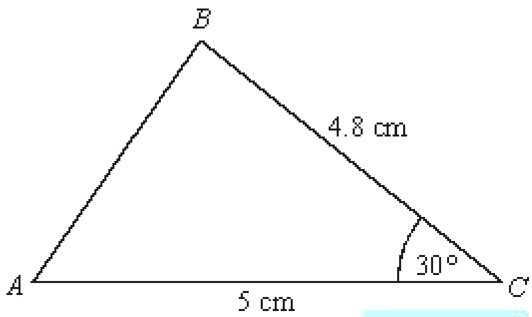
(ii) Hence calculate the area of the shaded segment to the nearest cm^2.

(3)

(Total 10 marks)

Q17.

The diagram shows a triangle ABC.



The lengths of AC and BC are 5 cm and 4.8 cm respectively.

The size of the angle BCA is 30°.

(a) Calculate the area of the triangle ABC.

(2)

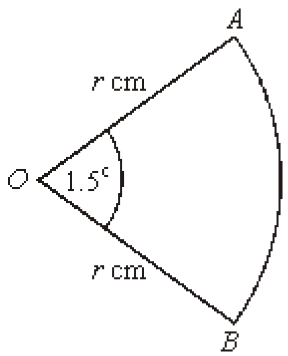
(b) Calculate the length of AB, giving your answer to three significant figures.

(3)

(Total 5 marks)

Q18.

The diagram shows a sector OAB of a circle with centre O and radius r cm.



The angle AOB is 1.5 radians. The perimeter of the sector is 56 cm.

(a) Show that $r = 16$.

(3)

(b) Find the area of the sector.

(2)

(Total 5 marks)